



Technical paper

Plasma arc welding: Process, sensing, control and modeling

C.S. Wu*, L. Wang, W.J. Ren, X.Y. Zhang

MOE Key Lab for Liquid-Solid Structure Evolution and Materials Processing, and Institute of Materials Joining, Shandong University, Jinan 250061, China



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ABSTRACT

This article introduces the basic principles of plasma arc welding (PAW) and provides a survey of the latest research and applications in the field. The PAW process is compared to gas tungsten arc welding, its process characteristics are listed, the classification is made, and two modes of operation in PAW, i.e., melt-in and keyhole, are explained. The keyhole mechanism and its influencing factors are introduced. The sensing and control methodologies of the PAW process are reviewed. The coupled behaviors of weld pool and keyhole, the heat transfer and fluid flow as well as three-dimensional modeling and simulation in PAW are discussed. Finally, a novel PAW process variant, the controlled pulse keyholing process and the corresponding experimental system are introduced.

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1. Introduction

Plasma is the state of the matter when part of the gas is ionized making it a conductor of electric current. It is the state of the matter present in between the electrodes in any arc [1]. Plasma arc welding is an arc welding process that melts and joins metals by heating them with a constricted plasma arc established between a tungsten electrode and the metals. The plasma arc welding (PAW) closely resembles the tungsten-inert gas (TIG) or gas tungsten arc (GTA) welding process in that it also uses a non-consumable tungsten electrode and a shielding gas such as argon. The tungsten electrode sticks out of the shielding gas nozzle in TIG (Fig. 1a) while it is recessed in the orifice gas nozzle in PAW (Fig. 1b). The main difference is in the construction of the torch. In plasma arc welding, the plasma arc is tightly constrained as shown in Fig. 1. A small amount of pure argon gas flow is allowed through the inner orifice surrounding the tungsten electrode to form the plasma gas. Because of the squeezing action of the constricting nozzle, the arc in PAW is concentrated and straight. This constriction increases the heat contained per unit volume of the arc plasma. Thus, arc temperatures of the order of 11,000 °C are not unusual in PAW.

To initiate the arc in PAW, a low current pilot arc is obtained between the electrode and the constricting nozzle first, which ionizes the plasma gas flowing through the nozzle. The plasma gas flowing through the constriction reaches a very high temperature, and provides a low resistance path to initiate the plasma arc

between the electrode and the workpiece. This is termed as a transferred arc. The transfer arc flows through the nozzle and extends its path to reach the base metal, forming a low resistance path of current between the tungsten electrode and the base metal. The plasma arc is with high temperature and velocity due to the restriction by the orifice, fusing and joining the metal pieces as desired. The plasma gas itself is not sufficient to protect the weld metal and therefore, a large volume of inert shielding gas is allowed to flow through an outer gas nozzle surrounding the inner nozzle, as shown in Fig. 1b. The shielding gases that can be used are argon, or helium.

There are two modes of operation in PAW, i.e., melt-in and keyhole. When energy is transferred to a workpiece from plasma arc the energy strikes the surface and immediately causes heating by a combination of conduction of heat in the arc and conversion of kinetic energy. If the rate at which energy is being deposited exceeds the rate at which heat is being conducted away, the temperature will rise to eventually cause melting and produce a fusion weld. This mode of energy deposition and weld production is called the melt-in mode or the conduction mode. If the density of the energy coming from the plasma arc is high enough, the rate at which it is deposited greatly exceeds the rate at which it is lost by being conducted into the workpiece. In this case, the plasma arc can penetrate through the molten pool and form a small throughout hole in the weld pool, referred to as keyhole. Moving the welding torch and the associated keyhole will cause the flow of the molten metal surrounding the keyhole to the rear region where it re-solidifies to form a weld bead. This is the keyhole mode in PAW [2]. Clearly, the keyhole mode produces larger penetration than the melt-in mode.

This article provides an introduction to the basic principles of plasma arc welding as well as a survey of the latest research and

* Corresponding author. Tel.: +86 531 88392711.

E-mail address: wucs@sdu.edu.cn (C.S. Wu).

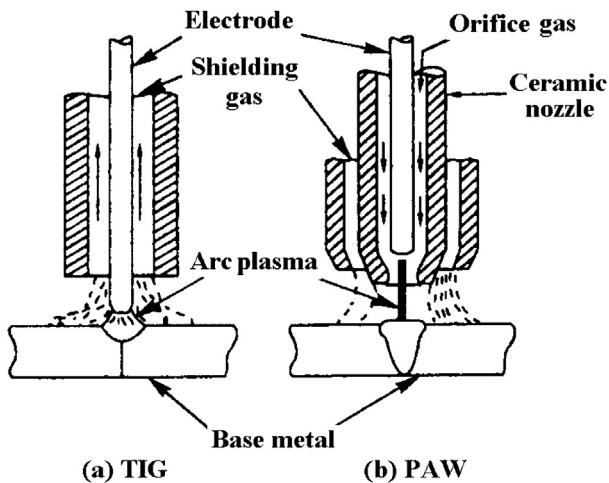


Fig. 1. Comparison of the torch construction of TIG and PAW processes.

applications in the field. First, the PAW process characteristics and keyhole mechanism are introduced. Then, the sensing and control of PAW process are reviewed. Following discussion of modeling and simulation work on PAW, the description of the PAW process variant is given.

2. PAW process

2.1. Characteristics

The arc used in PAW is constricted by a small nozzle and has a much higher velocity (300–2000 m/s) and heat input intensity (10^9 – 10^{10} W/m²) than that in conventional gas tungsten arc welding (GTAW) [3,4]. As a consequence, PAW has many advantages over GTAW, in terms of penetration depth, joint preparation and thermal distortion, etc. [5,6]. To be specific, PAW has following characteristics:

- (1) PAW offers greater welding speed, greater energy concentration and high efficiency than GTAW, which make it to be one of the most effective processes for many applications. And PAW gives better penetration than GTAW. Besides, welding in keyhole mode can make full-penetration welds in relatively thick material in a single pass.
- (2) PAW is an arc welding process that uses a non-consumable tungsten or tungsten alloy electrode. As shown in Fig. 2, recessing the electrode in a nozzle is an advantage that electrode contamination is minimized [7], which increases electrode service life. An electrode can usually last for an entire production shift without needing to be reground.
- (3) PAW offers a large tolerance to joint gaps and misalignment. Although the arc is constricted, the plasma column has a significantly larger diameter than the electron or laser beams. PAW

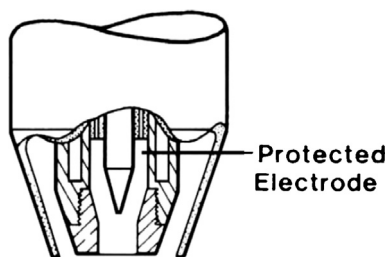


Fig. 2. Protected electrode.

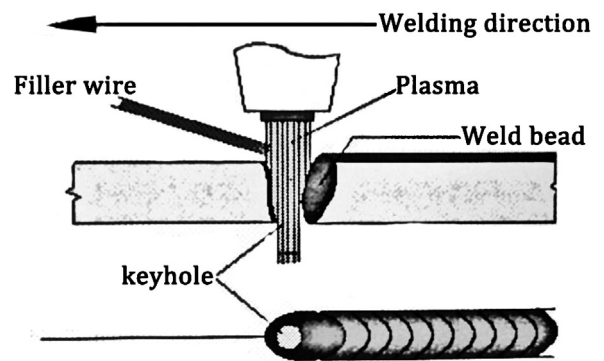


Fig. 3. Schematic of keyhole plasma arc welding.

minimizes the need for costly joint preparation and reduces or eliminates the need for filler metal.

- (4) The high depth-to-width ratio of a plasma weld compared to a GTAW weld also can greatly reduce angular distortion and residual stress. Narrower HAZ, less internal defects and better welding processing property make it an effective method to weld those structural components whose back-sides are difficult to be welded such as sealed container and small-diameter pipe [8].

However, although PAW has potential to replace GTAW in many applications as a primary process for precise joining [5], the equipment of PAW is more complex and costly, and the need for water cooling of the torch limits how small the torch can be made (GTAW torches may be gas-cooled and can be made to fit into smaller areas). Also, power supply and electric control systems are more complex than those in GTAW. Moreover, the adjustment and matching of its parameters are difficult, and it needs a high installation requirement of torch, nozzle, and tungsten.

2.2. Classification

As aforementioned, plasma arc welding is typically used in two modes of operation, melt-in fusion welds and keyhole fusion welds, on the basis of that whether the keyhole forms or not during the welding. The melt-in mode is accomplished with a softer, less constricted arc, using lower plasma gas flow rates, a minimum electrode setback, and current levels in the range of approximately 1–200 A [9]. This provides a slightly wider weld bead than when the arc is constricted [5], but in most cases allows for travel speeds equal to or greater than those typical with GTAW. The arc characteristics of this type weld mode are very similar to that of gas tungsten arc welding with additional advantages in many applications. This type weld mode is used for one-side welding with back formation of the thin plate or the double-side welding and multi-layer welding of the thick plate. Generally, when the current is lower than 30 A the melt-in fusion welding is called micro-plasma welding.

The keyhole welding is generally obtained by using a stiff and constricted arc. With increased plasma gas flow rate and electrode setback, a hole known as the “keyhole” is pierced through the entire metal thickness at the leading edge of the weld pool, where the forces of plasma column displace the molten metal, as shown in Fig. 3. Typically this technique is used for square butt welds on material thickness from 2.4 mm to 8 mm requiring 100% penetration in a single pass. Compared to laser welding and electron beam welding, keyhole PAW is more cost effective and more tolerant of joint preparation, though its energy is less dense and its keyhole is wider [4]. Thus, keyhole PAW has found applications on the welding of structural steels [10], automobiles [11], airplanes [12], rockets [13], space shuttles [14] and possibly on welding in space [8].

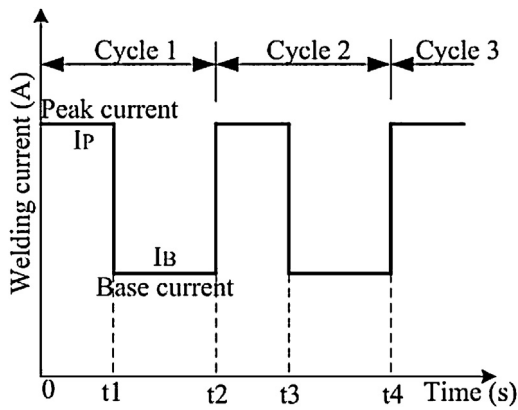


Fig. 4. Typical square-waveform of welding current.

The keyhole PAW includes several types as follows: DC plasma arc welding, pulsed plasma arc welding, variable polarity plasma arc welding, and other special plasma arc welding.

- (1) *DC plasma arc welding.* It is called DC plasma arc welding when the constant direct current is used in PAW. In the initial stage of the study on plasma arc welding, constant direct current is applied at first. The current in the DC PAW depends on the plate thickness and the penetration requirement. And the slight fluctuation of the welding parameters (e.g. welding velocity, plate thickness, welding current) will result in the closure of the keyhole and the instability of the welding process because of the thermodynamic equilibrium state depending on multitudinous factors. Consequently, proper range of the welding parameters should be chosen cautiously for the stable welding process in the DC keyhole PAW [15]. However, its application is limited because of the small regulation margin.
- (2) *Pulsed plasma arc welding.* During PAW process, keyhole is formed inside the weld pool, and it is sensitive to any changes of the welding parameters. Whether the keyhole condition is in quasi-steady state or not critically determines the stability of welding process and the quality of weld joints [16]. In conventional keyhole PAW, the appropriate parameter-window is narrower [17,18]. To expand the parameter-window for quasi-steady state of keyhole, pulsed current PAW is invented to produce one keyhole for each pulse with a pulse current waveform such as typical square-waveform of welding current [18], as shown in Fig. 4. After the keyhole is established, the current is switched from the peak current to the low base current so that the keyhole is closed to avoid burn-through [19].

The pulsed heating of the metal expands the adjustable range of the thermal conditions in welding and reduces heat accumulation on the metal. As a result, welding heat input can be controlled strictly to guarantee a good quality of the welding bead in all-position welding. However, for square waveform PAW, the mode “one keyhole in each pulse” is usually suitable to welding of metal sheet thinner than 4 mm [18].

- (3) *Variable polarity plasma arc welding (VPPAW).* Standard plasma arc welding, though used effectively on many other metals, had not been practical on aluminum alloys until the incorporation of the variable polarity characteristic. Variable polarity plasma arc welding, i.e., VPPAW process, combining the unique plasma keyhole capabilities with cyclic variable polarity, was developed by Van Cleave at Boeing Company [5], which provides many benefits for aluminum applications such as aerospace manufacture [14].

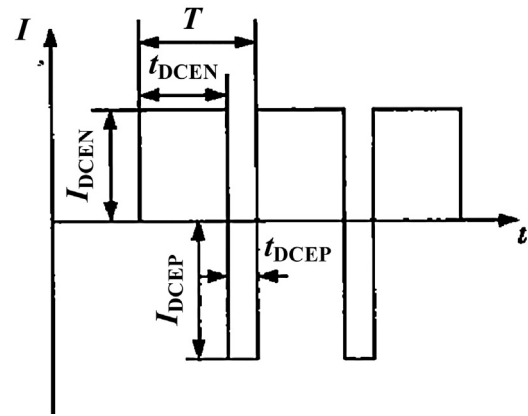


Fig. 5. Schematic of variable polarity current waveform.

Electrode negative operation (straight polarity) is necessary for plasma arc and it provides sufficient constant heat to the workpiece and minimizes heat buildup at the tungsten. But it would produce an unstable arc and irregularly shaped weld bead. Reverse polarity (electrode is positive and workpiece is negative) has the advantage that the workpiece is subjected to a cleaning process due to the phenomenon of cathode cleaning action, by the impingement of heavy (compared to electrons) positive ions on the workpiece surface and the evaporation of oxides due to high current density on the cathode spot.

As shown in Fig. 5, a variable polarity square-wave with unequal straight and reverse polarity times offers a combination of the high heating capability of straight polarity with the cleaning feature of reverse polarity [20]. Adequate cleaning is obtained by incorporating a relatively short (one-tenth to one-fifth the duration of the straight polarity current) pulse of reverse polarity current into the welding current waveform. Fig. 6 is the schematic of variable polarity plasma arc welding.

2.3. Process parameters

In plasma arc welding, the quality of the weld depends on a large number of factors, especially the physical characteristics of the material to be welded and the welding parameters to be used [21]. The welding process parameters must be carefully balanced to maintain the stability of the keyhole and the weld pool. Instabilities can result in weld discontinuities such as excessive penetration and incomplete penetration. The main parameters having a significant effect on welding quality include nozzle shape and size, tungsten electrode set-back, the distance from the nozzle to the workpiece, tungsten electrode material and size, welding current, the composition and flow rate of the plasma gas, welding speed, the composition and flow rate of the shielding gas, etc.

- (1) *Nozzle shape and size.* Nozzle shape and size is basic guarantee of a reliable and stable PAW process, and is also the most sensitive

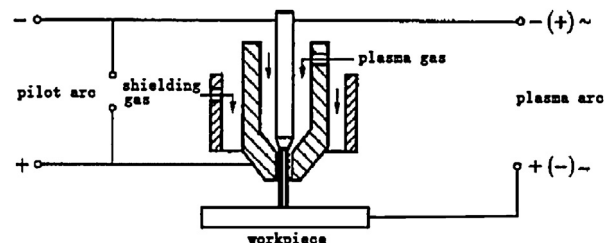


Fig. 6. Schematic of variable polarity plasma arc welding.

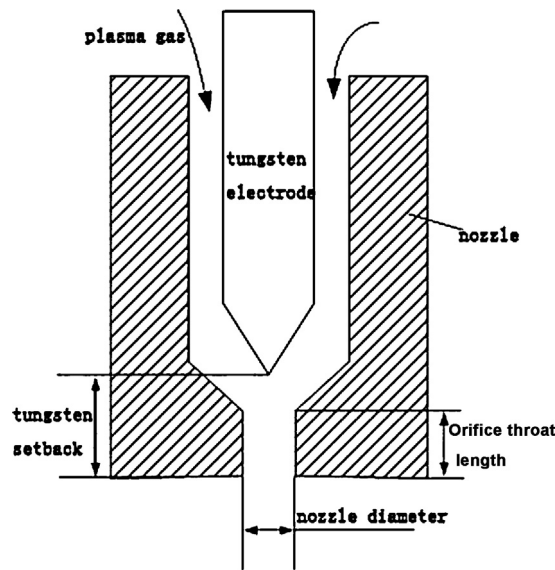


Fig. 7. Schematic of the nozzle of plasma arc welding.

factor affecting welding quality. Especially, the nozzle diameter and pore length are the most primary parameters, as shown in Fig. 7.

- (2) **Tungsten electrode material and size.** Tungsten electrodes are used when arc welding with the TIG and PAW process. In both processes the electrode, arc and weld pool are protected from atmospheric contamination by an inert gas. A tungsten electrode is used because it can withstand very high temperatures with minimal melting or erosion. Tungsten electrodes usually contain small quantities of other metallic oxides which can offer some benefits such as facilitating arc starting, increasing arc stability, improving current-carrying capacity of the rod, reducing the risk of weld contamination and increasing electrode service life.

Table 1 gives the corresponding permitted currents for different tungsten electrodes, which contain different metallic oxides [22]. Tungsten electrodes diameters depend on the welding currents. Higher current needs an electrode of larger diameter.

- (3) **Tungsten electrode set-back.** Tungsten electrode set-back refers to the distance between the end of the electrode and the nozzle exit, which usually is equal to or slightly larger than pore length, as shown in Fig. 7. This parameter has a significant effect on weld bead formation and arc penetration.
- (4) **Composition and flow rate of the plasma gas.** The plasma gas has direct effects through the plasma jet on the results of welding. The composition of the plasma gas and the its flow rate have a strong effect on the behavior of the weld pool, on the appearance of the weld, on the welding speed and on the weld defects [10]. Argon and argon–hydrogen mixtures are usually used as plasma gases.
- (5) **Welding current.** The welding current affects weld quality through the arc pressure and arc temperature. Studies show that an increased welding current widens the weld on both the surface and root sides [10]. An inadequate current level causes

Table 1
Permitted currents for tungsten electrodes of different component [22].

Type	Permitted current (A/mm ²)
Pure tungsten	26
Thoriated tungsten	44
Lanthanum tungsten	36
Yttrium tungsten	50

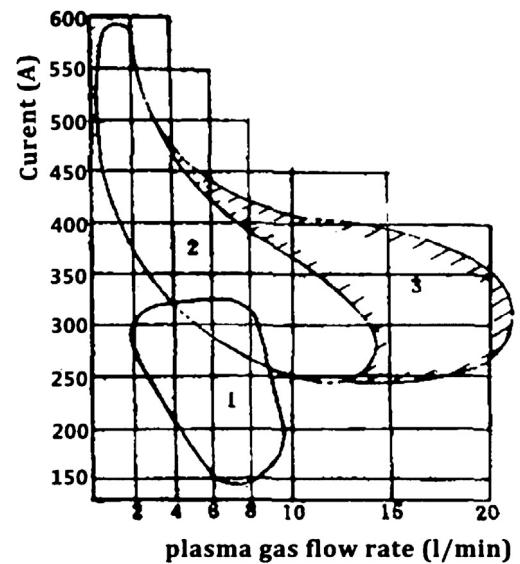


Fig. 8. Relationship between welding current and plasma gas rate [23] (1, cylinder orifice; 2, diffusing orifice; 3, region where undercut can be avoided by filling metal).

lack of penetration and slight undercutting in the toe area of the weld. Excessive current levels cause electrode damage either immediately at the start of welding or later [22].

The effect of welding current on weld quality is nearly always connected to the welding speed and the plasma gas type and flow rate. The formation of the weld and the mechanical properties of the weld joint will be best if a low plasma gas flow rate and a high welding current are used. At the same time, the welding speed will be high, which is also most economical. Fig. 8 shows the matching relationships of keyhole plasma arc welding parameters.

- (6) **Welding speed.** As shown in Fig. 9, there is a wide range of variation in welding speed to produce acceptable welds, if the welding current is suitable. Of course too high welding speed, in relation to the other parameters especially the welding current, will result in undercutting and incomplete penetration occurs. On the contrary, inadequate welding speed leads to excessive penetration. It is desirable to weld at the highest possible welding speed for the sake of economic.
- (7) **Distance from the nozzle to the workpiece.** In the keyhole PAW, the distance from the nozzle to the workpiece is not a very sensitive parameter because of high arc stiffness. But too small height will lead to narrow weld and undercut, and too high one will affect the arc penetration and the welding quality. Generally, appropriate distance is 3–12 mm.

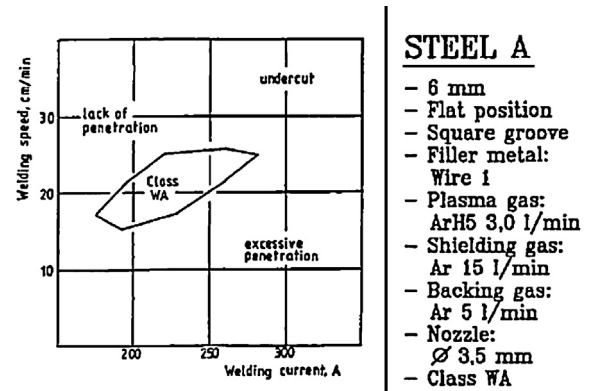


Fig. 9. Relationship between welding current and the welding speed [10].

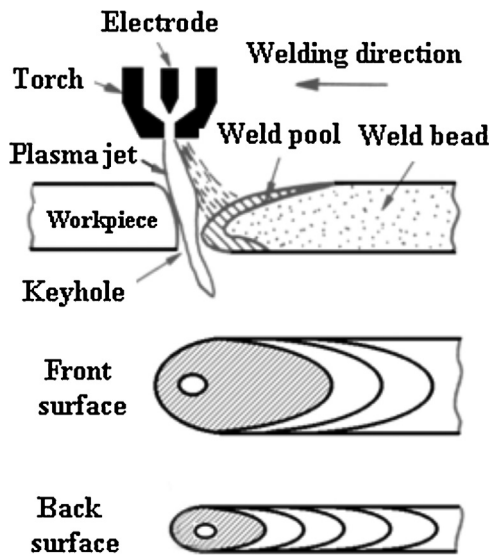


Fig. 10. Schematic diagram of keyhole PAW process.

- (8) *Composition and flow rate of the shielding gas.* The effect of shielding gas on the weld quality of plasma arc welding is not as important as that of the plasma gas, but it cannot be ignored entirely. The effects are particularly noticeable in welding of joint roots. The possible shielding gases are argon, argon–hydrogen, argon–carbon dioxide and argon–oxygen gas mixtures.

3. Keyholing process

The plasma arc torch has a water-cooled copper nozzle in front of the cathode to constrict the arc plasma through thermal pinch effect. So the arc plasma with high energy density and high arc pressure can be produced. As shown in Fig. 10, when the plasma arc impinges on the area where two workpieces are to be joined, it can melt material and create a molten liquid pool (weld pool). Because of its high velocity and the associated momentum, the plasma arc can penetrate through the molten pool and form a keyhole in the weld pool. Moving the welding torch and the associated keyhole will cause the flow of the molten metal surrounding the keyhole to the rear region where it re-solidifies to form a weld bead [24].

In practice, the relative motion between the plasma jet and workpiece is often held at zero until full keyholing is obtained [25]. During the keyhole formation process (keyholing), the keyhole undergoes the following stages:

- (1) *Blind keyhole.* A cavity is formed inside the weld pool, but it has not completely penetrated through the workpiece.
- (2) *Unstable open keyhole.* The cavity in the weld pool has been transformed into the penetrated keyhole. Because the open keyhole is just established, it is unstable and may close momentarily during a very short period.
- (3) *Open keyhole in quasi-steady state.* The open keyhole is consistently maintained. Meanwhile, the keyhole size just varies within a limited scope.

The size and shape of the keyhole produced by PAW depends mainly by the pressure of the impinging plasma jet [3,10,25], rather than the recoil pressure from the vaporization like the case in laser and electron beam welding.

In keyhole PAW, the quality of the weld depends on the keyhole stability, which depends on the heat transfer and fluid flow phenomena inside the weld pool. However, many factors make

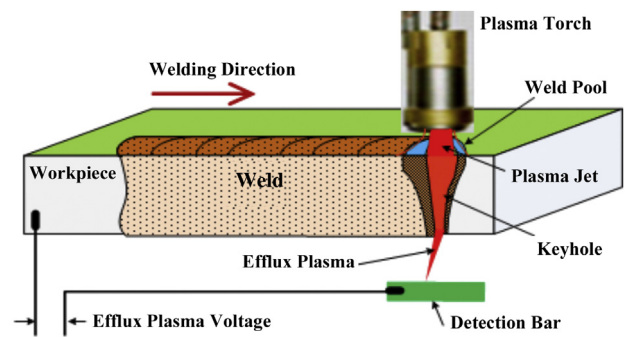


Fig. 11. Sensing of efflux plasma voltage.

effect on the stability of the keyhole. The fluid flow and heat transfer phenomena are very complicated in the keyhole welding where the weld pool is coupled with the keyhole and both interact with each other dynamically.

The dominating force tending to form and maintain the keyhole is the plasma stream pressure. The recoil pressure exerted on the weld pool surface due to the evaporating particles leaving the surface plays a minor role in PAW [3,10,25]. The forces tending to close the keyhole include the gravitational pressure and the surface tension. The gravity at any point in the liquid surrounding the keyhole is directly proportional to the density and height of metal above it. The effect of surface tension is the force which tends to minimize the surface area of the molten metal. In keyholing, before workpiece breakthrough surface tension tends to act upwards against the arc plasma pressure and recoil pressure.

Inside the weld pool surrounding the keyhole, the pressure of plasma arc, surface tension, buoyancy, and electromagnetic force are four main forces determining the fluid flow and heat transfer.

4. Sensing and control of keyhole status

In keyhole plasma arc welding, the stability of keyhole is an important factor that affects the stability of the welding process and the quality of the weld joint. Thus, monitoring, sensing and control of keyhole status are very important to achieve defect-free and high-quality welds. Various measuring methods have been tried to detect the keyhole condition in PAW process.

4.1. Sensing the efflux plasma voltage signal

As shown in Fig. 11, when an open keyhole is established, the efflux plasma will exit through the keyhole. A piece of metal sheet is chosen as the detection bar mounted underneath the workpiece to be welded and kept insulated. Due to the phenomenon of plasma space charge, the efflux plasma establishes an electrical potential between the workpiece and the detection bar. However, if the open keyhole is not established or closed because of insufficient heat and force actions, there is no efflux plasma between the workpiece and the detection bar and thus no electrical potential is established. Hence, the electrical voltage resulted from the plasma efflux can be measured to characterize the establishment of keyhole and the keyhole size [26–29]. The correlations of welding current, efflux plasma voltage and backside weld width are illustrated in Fig. 12. During the PAW process, increasing of welding current within some range causes stronger heat and force action from the arc plasma, so that the keyhole dimension grows as welding current increases. The growing of keyhole dimension results in increasing of efflux plasma voltage and wider weld bead at backside.

But this is just an indirect approach correlating the electrical potential amplitude with the intensity of efflux plasma exiting from the keyhole.

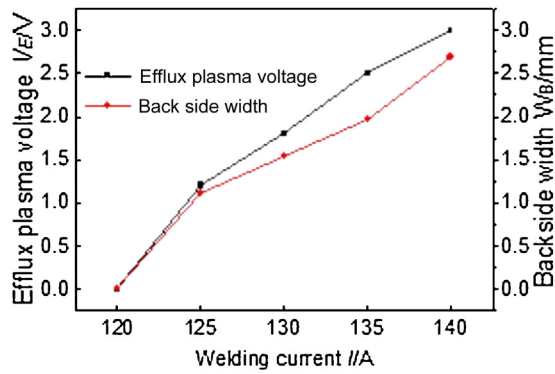


Fig. 12. Correlations of welding current, efflux plasma voltage and backside width.

4.2. Sensing the sound signal

Under different process conditions in variable polarity PAW, such as no-keyhole (melt-in), keyhole and cutting, the corresponding weld pool produces different acoustical signal which may be detected to classify the three modes of weld pool [30]. As shown in Fig. 13, the arc sounds were detected by using a microphone for identifying the no-keyhole mode, transition mode and keyhole mode generated from a work-piece with a variable thickness [31]. However, the reliability of sound signal based sensors is not satisfactory because of the interference from environmental noises.

4.3. Sensing the light signal

Investigators at NASA used spectroscopic analysis of the plasma arc light to monitor the keyhole [32]. It was found that the relative difference between the spectral line strength of hydrogen and argon atoms could reflect whether the keyhole was formed or not. But it was difficult to distinguish the stable keyhole status from the weld cutting status.

As shown in Fig. 14, the plasma arc light was once detected by spectroscopic approach to obtain information concerning the keyhole formation and collapse [33]. But the detection signal cannot accurately reflect the size of the keyhole in real time.

4.4. Plasma cloud charge sensor

To detect the state of the keyhole reliably, Zhang et al. developed sensors to monitor the plasma reflection [34,35]. When the keyhole is not fully penetrated (blind keyhole), the plasma arc has to be reflected from the cavity. After the keyhole is fully established (open keyhole), the plasma jet will exit at the bottom of the penetrated keyhole, so that the amount of reflected plasma, if any, will be significantly reduced. Hence, a copper probe [34] or camera [35] was used to monitor the establishment of the penetrated keyhole based on the behavior of the reflected plasma. As shown in Fig. 15, the plasma cloud charge sensor monitors the charge effect of the

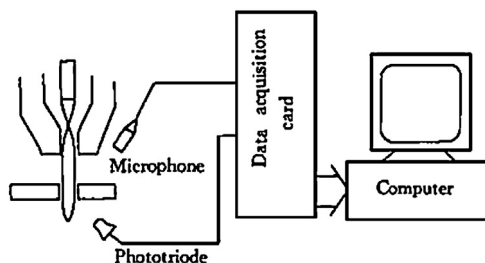


Fig. 13. Schematic of sound signal sensing in welding process [31].

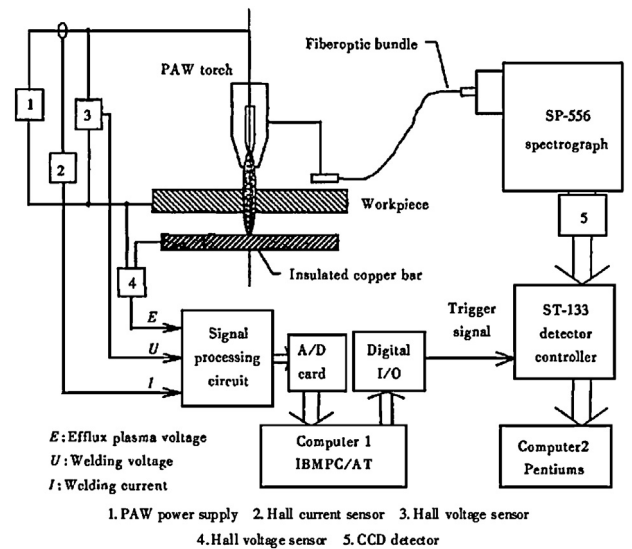


Fig. 14. Schematic of arc light sensing of keyhole behavior [33].

dynamic change of the plasma cloud above the surface of the work piece. The voltage between the probe and the workpiece can reflect the angle of plasma cloud which is an indicator of keyhole status. However, such sensing approach is also indirect, and is unable to provide quantitative information on the shape and size of an open keyhole.

4.5. Visual image sensing

Vision sensor can provide more direct information on the keyhole condition and size. An ultra-high shutter speed vision system was used to simultaneously image the keyhole and the weld pool from the backside of the work-piece [36]. As shown in Fig. 16, the camera system includes a strobe-illumination unit (pulse laser), camera head and system controller. The pulse duration of the laser was 3 ns, and the camera was synchronized with the laser pulse, thus, the intensity of laser illumination during the peak was much higher than that of the plasma efflux. Both the keyhole and the weld pool could be imaged clearly and simultaneously, as shown in Fig. 17. But such a laser-strobe system is complicated and very expensive in practical applications.

A low-cost visual sensor is more attractive for sensing keyhole in PAW process. To this end, Wu's team developed a cost-effective vision system for observing the keyhole from the backside of

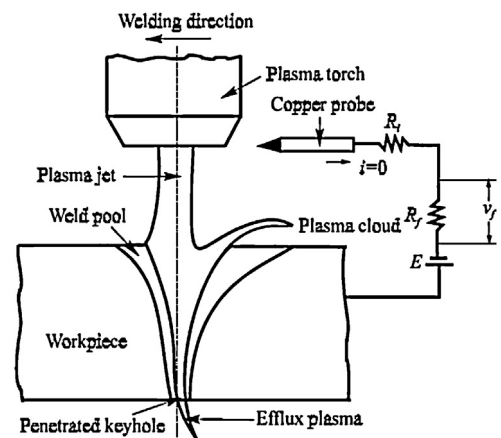


Fig. 15. Schematic of plasma cloud sensing [34].

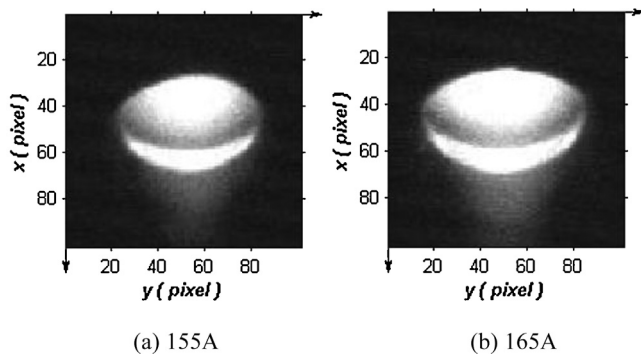


Fig. 19. The raw images of keyhole under different level of welding current [37]: (a) 155 A and (b) 165 A.

example, the sampled value of the efflux plasma voltage was taken as the control variable to describe the keyhole status, while the pulse current value and its dropping slopes were taken as the controlling variables [29]. Some special control algorithms are developed to establish the correlation between the controlling variables and the controlled variable, such as the proportional and integral control algorithm [29], the predictive control algorithm [19], the interval model control algorithm [19,39], and nonlinear interval model control algorithm [40].

5. Modeling and simulation

The formation and stability of keyhole as well as the weld pool behaviors surrounded it are related to complicated phenomena, such as the dynamic balance of various forces acting on the surfaces of keyhole and weld pool, fluid flow and heat transfer, melting and solidification. All these physical phenomena have decisive effects on the weld quality. To implement the keyhole PAW successfully, it is essential to achieve full understanding of the process mechanism related to all these physical phenomena. Therefore, it is of great significance to model and simulate the dynamic development of keyhole and associated physical behaviors in keyhole PAW process.

Because of the complexity of the phenomena associated with the formation of a keyhole, only a limited number of theoretical studies treating the PAW process have been reported, each of varying degrees of approximation, and each focusing on different aspects of the problem. Hsu and Rubinsky [41] performed a study on the two-dimensional, stationary thermal phenomena during the heating of a stainless steel plate with a predrilled hole by penetrating plasma arc, which does not rigorously simulate a welding process. They also conducted the calculation for the two-dimensional (in the plane normal to the plasma jet flow axis) weld pool flow and temperature fields associated with steady travel PAW, but neglect surface tension effects and simplify the pool geometry by assuming flat upper and lower free surfaces and a keyhole of constant radius, and a non-tapering solid–liquid phase boundary [42]. Nehad established a model for transient two-dimensional heat transfer problems during PAW process with assuming a circular shape of keyhole and neglecting the metal vaporization at the plasma–liquid interface [43]. Keanini and Rubinsky [8] conducted an investigation focusing on surface tension phenomena to predict the interfacial shape between a plasma jet and a liquid metal in the case of a plasma arc jet penetrating through a stationary metal plate. Fluid motion within the molten metal was neglected and the jet was modeled as a one-dimensional compressible ideal gas flow. The shape of the plasma jet–liquid metal interface was determined using the Young–Laplace equation, and then the position of the liquid–solid phase boundary was roughly estimated

by neglecting convection and axial conduction, and by assuming that the liquid at the plasma interface was at its boiling point, and the liquid within the melt behaves as a static thermal boundary layer.

The most comprehensive model of PAW to data is Keanini and Rubinsky's quasi-steady-state three-dimensional finite element simulation to calculate the weld pool shape, the pool's fluid flow and temperature fields [44]. However, the capillary surface shapes were determined by performing one Newton–Raphson iteration on the initial surface guess which was based on experimentally observed shapes, i.e., the keyhole shape was assumed according to experimental results [44]. Because of these somewhat oversimplified assumptions, the calculated weld pool shape, particularly the vertical cross sections at the rear portion of the weld pool, are not in agreement with the expected ones. Fan et al. [45] developed a two-dimensional model to demonstrate the heat transfer and fluid flow in stationary keyhole PAW.

On the other hand, from the viewpoint of engineering applications, emphasis is put on developing a suitable heat source model that reflects the thermo-physical characteristics of keyhole PAW process, and then the numerical analysis of temperature field in keyhole PAW has been undertaken to obtain the correlation of process parameters with the weld dimension and geometry [46,47]. However, this methodology oversimplified the complicated physical mechanisms during keyhole PAW process.

Wu's team considered the coupled behaviors of weld pool and keyhole to develop a three-dimensional model for analyzing the heat and fluid flow inside a weld pool with a dynamic keyhole [48–51]. In view of the characteristics of PAW process, a combined volumetric heat source model (double-ellipsoid plus conic body source) is established, and one of its distribution parameters is adjusted dynamically with the variation of the depth of keyhole. The physical phenomena, such as the weld pool development, the keyhole formation, the evolution of fluid flow and thermal field, the full-penetration of the test plates, and the transformation from a blind keyhole to an open keyhole, etc., are quantitatively analyzed. The numerical results reveal the regularity of fluid flow in weld pool with a keyhole. The calculated keyhole shape and the fusion zone of plasma arc welds are compared with the experimental measurements. Both agree with each other generally. It lays foundation for optimizing the welding process parameters and improving the stability of plasma arc welding process. Fig. 20 shows the evolution of keyhole shape, fluid flow and temperature profiles in weld pool under the condition of a test case [50].

Fig. 21 shows the keyhole sizes versus time for two test cases. It can be seen that the keyhole evolution process from its emergence to expansion does not occur at a uniform speed. At the initial stage before 0.4 s, the keyhole depth rises slowly, but its topside width increases quickly. During the period of 0.4–0.6 s, the depth rises markedly, while the width rises slowly. When an open keyhole is formed and its quasi-steady state is reached, its width keeps almost constant. Comparing test case 2 with case 3, it can be seen the increscent slope of keyhole depth is steeper in test case 3 than that in test case 2 during the period of 0.4–0.6 s when the keyhole transforms from blind to open state. However, for the keyhole width curves, the slopes are not changed so much.

A keyhole model is also established according to the force-balance condition on the keyhole wall [52,53]. The shape and dimensions of quasi-steady state keyhole are calculated. Fig. 22 shows the 3-D keyhole under different levels of welding current on stainless steel test-piece of thickness 8 mm. When the welding current is lower, the keyhole is not breakthrough, and blind keyhole appears (Fig. 22a and b). A current value of 150 A just produces an open keyhole (Fig. 22c). If the welding current is 160 A, a larger complete keyhole exists.

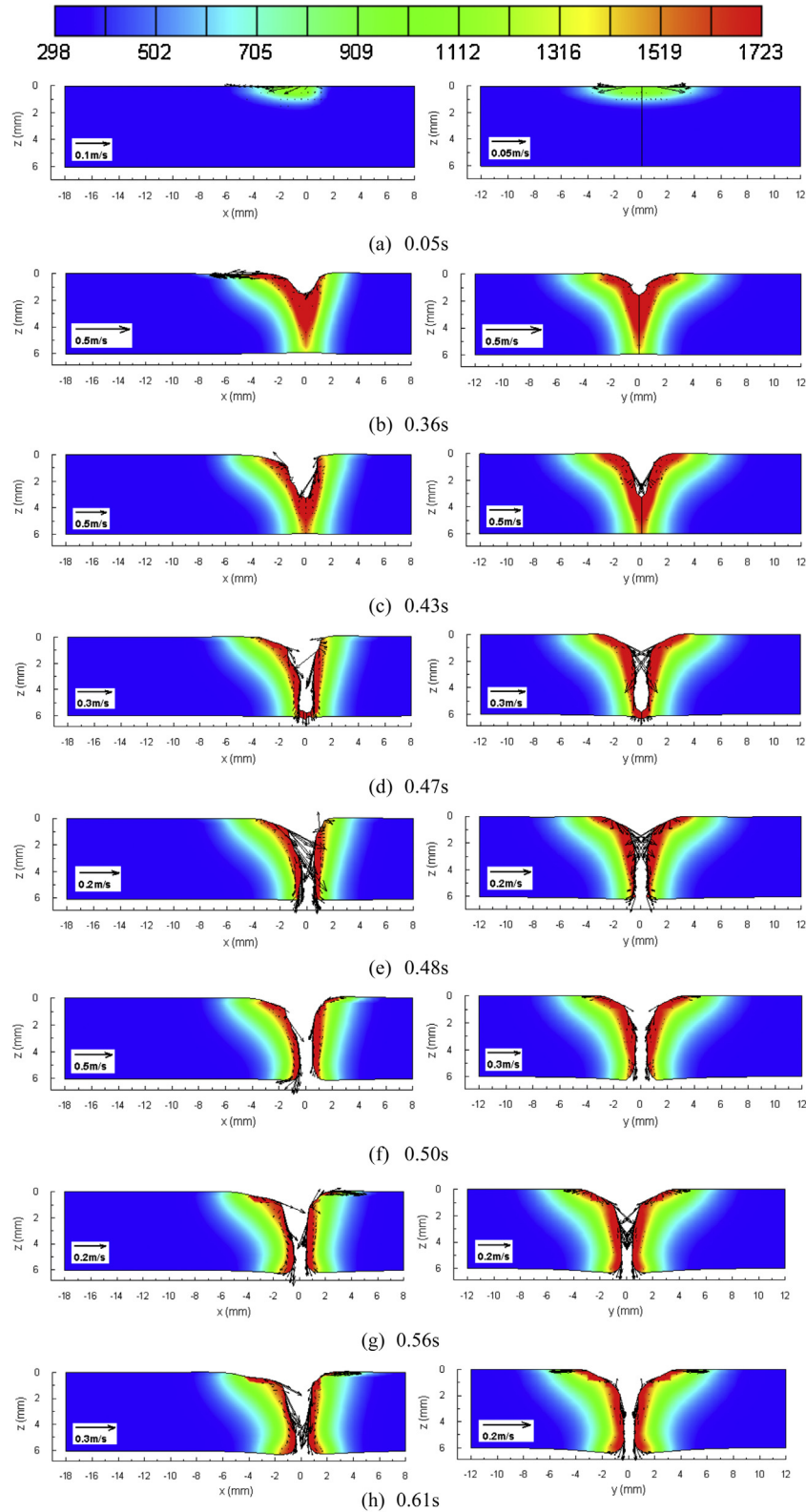


Fig. 20. Evolution of keyhole shape, temperature and fluid velocity profiles in weld pool (Test case 3) [50] (left: longitudinal cross-section; right: Transverse cross-section). (a) 0.05 s, (b) 0.36 s, (c) 0.43 s, (d) 0.47 s, (e) 0.48 s, (f) 0.50 s, (g) 0.56 s and (h) 0.61 s.

6. Variant-controlled pulse keyholing PAW

In normal keyhole PAW, the keyhole is maintained open and stable to ensure weld quality. To this end, the welding process parameters, especially the welding current, have to be within a

narrower range [16]. If the welding current is a little bit lower, the keyhole may be closed, but if the welding current is a little bit higher, the volume of keyhole expands too large and there is a tendency of the molten metal being blown away from the weld pool, causing burn-through or cutting instead of joining. To overcome the

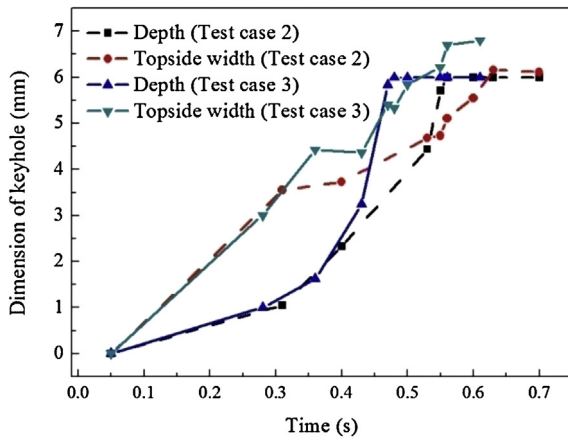


Fig. 21. Keyhole dimensions versus time [50].

shortcomings of conventional PAW process, the ‘controlled pulse key-holing’ strategy is proposed and the experimental system is developed [28,29].

As shown in Fig. 23, the keyhole sensor is developed through measuring the efflux plasma voltage. A simple sensor consisting of a resistor and a capacitor is used to detect the electrical potential or the voltage between the workpiece and the measuring bar. The larger the keyhole size, the higher the intensity of the efflux plasma, thus, the larger the efflux plasma voltage. There is a correlation of

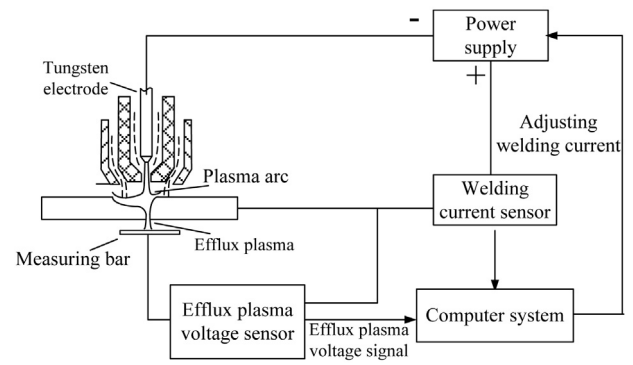
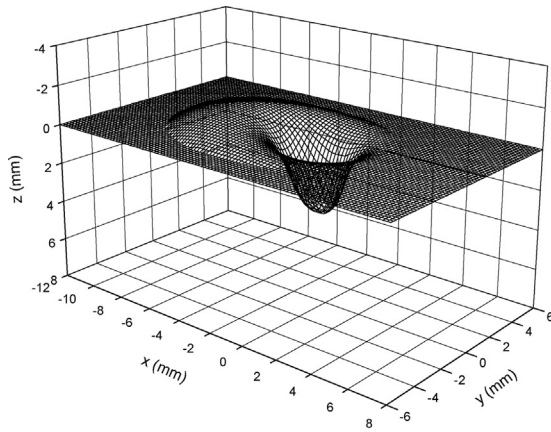


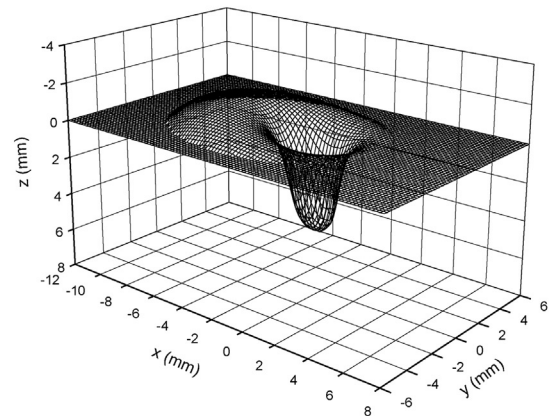
Fig. 23. The experimental set-up of the controlled pulse keyholing PAW [29].

the keyhole diameter at backside with the measured efflux plasma voltage signal [29].

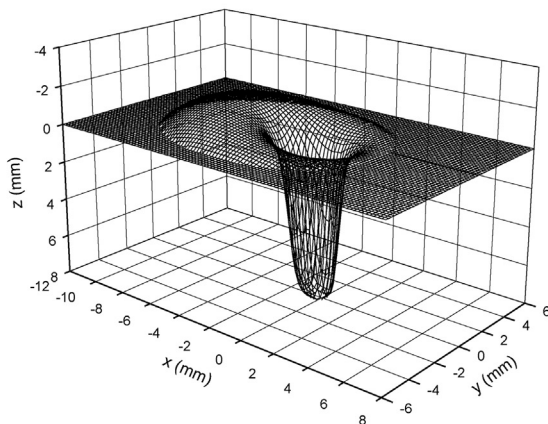
Fig. 24 shows the specially designed welding current waveform and keyhole signal. At the dropping stage of welding current from the peak level to the base level, two sub-stages of current decreasing with different slopes of K_1 and K_2 are added. At instant t_1 , the peak current I_p is applied, and the system keeps detecting the signal from the efflux sensor. The signal of the efflux plasma voltage is around zero before the keyhole is established. At instant t_2 , the keyhole is established and the efflux plasma voltage exceeds a certain value. To avoid the keyhole size expands too much, the



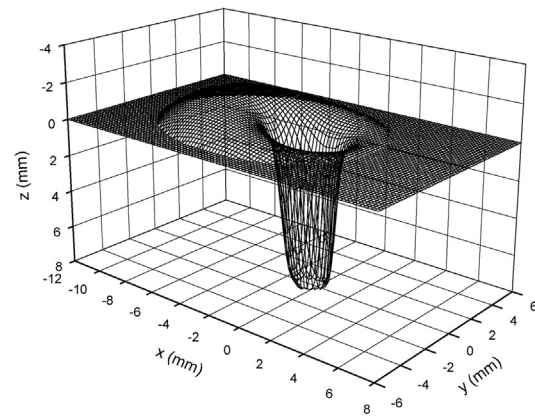
(a) 140A



(b) 145A



(c) 150A



(d) 160A

Fig. 22. Three-dimensional keyhole under different levels of weld current: (a) 140 A, (b) 145 A, (c) 150 A and (b) 160 A.

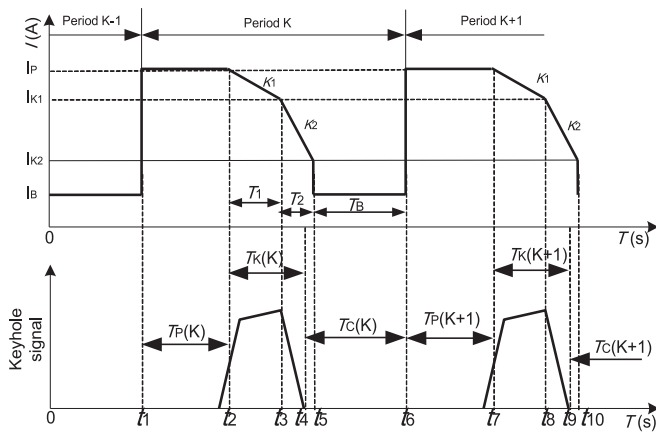


Fig. 24. The controlled pulse current waveform and keyhole signal [29].

current starts to decrease with a dropping slope K_1 . Because of thermal inertia, the keyhole volume remains to expand slowly even though both heat input and arc force associated with the welding current start to drop during this stage. At instant t_3 , when the keyhole size reaches the pre-set value to meet the desired practical requirements of weld quality, the current decreases at a steeper slope K_2 ($|K_2| > |K_1|$), so that the keyhole stops expanding but starts to close. At instant t_4 , the keyhole is completely closed, and the efflux plasma voltage is zero. The current is switched to base level I_B at instant t_5 . After I_B is applied for a pre-selected period $T_B = t_6 - t_5$, the current is switched to the peak level I_P again to begin a new cycle. In this way, it can ensure the keyhole establishing and full penetration but avoid burn-through defects.

To characterize the keyhole size and weld bead width at backside of the workpiece, the average value (V_{EP}) of efflux plasma voltage (V_E) from t_2 to t_3 in each cycle is used as the feedback signal because it has been validated that the measured V_{EP} is well correlated with the weld bead width at backside of the workpiece [28]. Take the average value of efflux plasma voltage (V_{EP}) as the controlled variable. Take the pulse current value (I_P) and its two dropping slopes (K_1 and K_2) as the controlling variables. Through adjusting I_P and K_1 & K_2 , keep V_{EP} within the preselected range so that the weld penetration and weld bead width is controlled. The PI (proportional and integral) controller is used to adjust I_P and K_1 & K_2 according to the difference between the measured V_{EP} and setpoint V_{EP} .

To test the control effectiveness of the developed control system further, a test plate with varied thickness is used, as shown in Fig. 25. Because the plate thickness is gradually varied from 8 mm to 4 mm, the welding process cannot maintain consistence if no appropriate control action is applied. By employing the developed system, the peak current and both its duration and dropping slopes are able to make suitable adjustment based on the measured efflux plasma voltage signal as the thickness varies along the welding direction. Fig. 26 is the photograph of the weld appearance at both topside and backside. It looks that the weld formation and penetration are controlled very well, even if the plate thickness changes so much.

To optimize this process variant-controlled pulse keyholing PAW, theoretical and experimental work has been conducted to

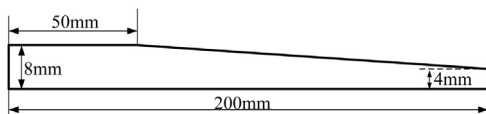


Fig. 25. The test plate with varied thickness [29].

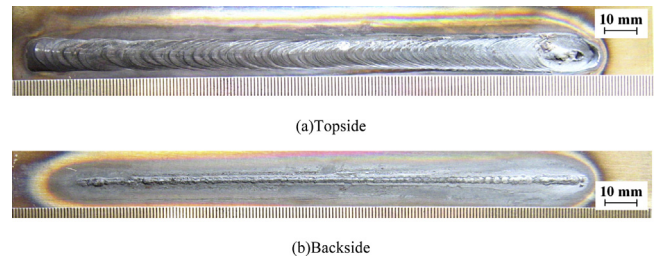


Fig. 26. The weld appearance at topside (a) and backside (b) of varied-thickness workpiece [29].

understand its underlying physical mechanisms [54–56]. However, further investigations are still needed to quantitatively analyze the interactions between the weld pool and the dynamic keyhole as well as the fluid flow and heat transfer inside the weld pool.

7. Summary and future work

In this review article, the state-of-the-art development in sensing, monitoring, modeling and control of plasma arc welding (PAW) processes is introduced. The technical characteristics and advantages of PAW process are discussed. The main process parameters, two modes of operation in PAW (melt-in and keyhole), and the classification are given out. The keyhole mechanism and keyholing process are described. Then, various sensing, measuring and control methods of the keyhole status are reviewed. To seek deep insight into the process mechanism, mathematical modeling of the PAW process is essential. The modeling and simulation progress on the keyhole evolution and its interaction with the weld pool is surveyed, and some examples of the predicted heat transfer and fluid flow in PAW are demonstrated. Finally, the controlled pulse keyholing PAW process is introduced.

Because PAW process is non-linear, dynamic and complex systems which has multiple inputs, multiple outputs, and multiple disturbances, an output is typically affected by more than one input or may even be affected by another output [57]. This coupling characteristic and the involved large number of process parameters require that multiple sensors be integrated to monitor multiple variables during the welding processes. Vision-based imaging of keyhole and weld pool should be combined with the sensing of the electrical signals (such as the efflux plasma voltage). The information from multi-sensors should be integrated and fused to describe the dynamics of both weld pool and keyhole so that weld quality is effectively controlled. Furthermore, modeling and simulation of the coupled fluid flow and heat transfer in both the plasma arc and the weld pool should be conducted to deal with the interaction between the plasma arc and keyhole wall inside the weld pool.

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